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3.4: Research Publications and Awards

3.4.3: Number of research papers published per teacher in the Journals as notified on UGC CARE list during the last five years

Research Journal Paper published in Year Jan-Dec 2022

Sr. No	Research Paper Title	Author Names	Link to Paper
1	Prediction of Dengue Outbreaks in Mumbai Region Based on Disease Surveillance and Meteorological Factors using Big Data Approach	Dr. D. R. Kalbande, Asha Bharambe	https://ijirt.org/master/publishedpaper/IJIRT154075_PAPER.pdf
2	Knowledge graphand semantic web model for cross domain	Shital Kakad , Sudhir Dhage	https://www.jatit.org/volumes/Vol100No16/25Vol100No16.pdf
3	Compact Planar Electromagnetic Bandgap Structure for Signal and Power Integrity Improvement in High-speed Circuits	Manisha R. Bansode, Surendra S. Rathod	https://www.jpier.org/ac_api/download.php?id=22102205
4	Dual Degree of Freedom Solar Tracking System	Devang Patani, Urmila Patil, Nayana Shelke, Dr. Rajendra Sutar, Prof. Najib Ghatte	https://ijaem.net/issue_dcp/Dual%20Degree%20of%20Freedom%20Solar%20Tracking%20System.pdf
5	Analytical modeling and numerical investigation of a variable width piezoresistive multilayer polymer micro-cantilever air flow sensor	P. V. Kasambe, Kiran S. Bhole, A. A. Bage, N. R. Raykar, D. V. Bhoir	https://www.tandfonline.com/doi/abs/10.1080/2374068X.2022.2076974?journalCode=tmpt20

6	Privacy-Preserving Mechanism to Secure IoTEnabled Smart Healthcare System in the Wireless Body Area Network	Renuka S. Pawar, Dhananjay R. Kalbande	https://www.ijcna.org/Manuscripts/IJCNA-2022-O-60.pdf
7	Optimization of quality of service using ECEBA protocol in wireless body area network	Renuka Pawar, Dhananjay Kalbande	https://link.springer.com/article/10.1007/s41870-022-01152-z
8	QOE Improvement for Dynamic Adaptive Streaming of Multimedia in LTE Cellular Network Using Cross-layer Communication	Dr. Anand D Mane, Dr. Uday Pandit Khot	https://www.mecs-press.org/ijwmt/ijwmt-v12-n2/IJWMT-V12-N2-5.pdf
9	Ultrasound Image Enhancement using Super Resolution	Ashwini S Sawant ,Sujata Kulkarni	https://www.sciencedirect.com/science/article/pii/S2667099222000159
10	Helping Hand: Metro Travel Assistance App for Elderly and Differently Abled People	Prof. Harshil Kanakia ,Pallavi Thakur,Vishal Parab,Shivam Chaubey,Manish Jha	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4100953
11	Mood Detection using Sentiment Analysis	Neil Khanolkar, Ajinkya Sathe, Ketaki Shinde, Aarti M. Karande	https://www.ijcaonline.org/archives/volume184/number26/32475-32475-2022922316
12	The Need of Explainability to Enhance the Trustworthiness of Decision-Making System	Aarti Karande, Prachi Dalvi, Harshal Dalvi, Milind Karande	https://doi.org/10.37896/HTL28.07/6190
13	Electron emission from water vapor under the impact of 250-keV protons	Abhijeet Bhogale, S. Bhattacharjee, M. Roy Chowdhury, Chandan Bagdia, M. F. Rojas, J. M. Monti, A. Jorge, M. Horbatsch	https://journals.aps.org/pra/abstract/10.1103/PhysRevA.105.062822#:~:text=Absolute%20double%20differential%20cross%20sections,by%20using%20hemispherical%20electrostatic%20analyzer.

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Prediction of Dengue Outbreaks in Mumbai Region Based on Disease Surveillance and Meteorological Factors using Big Data Approach

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Abstract - **One of the most prevalent vector-borne disease in India is Dengue. This study investigated the effects of various environmental/climatic factors on dengue incidence using time series analysis and machine learning approach using the big data environment. The aim of the current study is to understand environmental impact on vector-borne disease (Dengue) in a particular area from Mumbai and use it for prediction of the disease. Disease Incidence Data for dengue was collected from 2012 to 2019 from the study hospital in Mumbai while the climatic data was obtained from the web resources. Statistical time series analysis methods and machine learning techniques are used to make predictions. Our preliminary analysis has identified that there is a correlation between Dengue incidence rate and climatic conditions. Amongst machine learning model, random forest technique gave the results with RMSE of 47.08. Amongst time series analysis techniques, SARIMA model gave significant results with RMSE as 20.79 in univariate analysis as compared to ARIMA with RMSE of 61.27. The multivariate VAR model gave results with RMSE of 27.32. In both univariate and multivariate models, we concluded that when confounding factors are incorporated into forecasting model, it significantly improves AIC value (an AIC value of 5.89 was obtained for VAR model).**

Index Terms - **Big Data, Dengue, Machine Learning, Prediction, Time-Series.**

I.INTRODUCTION

Dengue transmission predominantly occurs in tropical and subtropical areas where mosquitoes can survive and multiply. Currently there are more than 100 countries that are endemic to dengue disease [1], [2] with about 390 million dengue virus infections per year [3] as shown in Figure-1 and 40% of world's population, about 3.9 billion people [2] are at the risk of suffering from the disease.

Figure 1: Prevalence of Dengue across world from 2010-2016. [3]

The Indian government has identified Dengue as important diseases to track from 12 diseases in laboratory confirmed form and from 22 diseases in presumptive form [4]. Hence a nationwide surveillance called Integrated Disease Surveillance Project (IDSP) [4] is conducted to identify these disease incidences.

Though nationwide health plans have succeeded in reducing fatality of few vector-borne disease to a certain extent, there is however a need for further improved and innovative dengue control measures [5] and efficient area-specific approach to predict the extent of these diseases. There are very few prediction systems developed so far to predict the spread of vector borne diseases in Mumbai city taking into considerations its environmental, social, and economic factors. Further there is no integrated data (clinical, disease and its environmental factors) available to model the disease.

Dengue is spread by the day-biting Aedes Aegypti mosquito that breeds in fresh water stored in

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KNOWLEDGE GRAPH AND SEMANTIC WEB MODEL FOR CROSS DOMAIN

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ABSTRACT

The industry and institutes systems are designed to develop a smart world. The heterogeneous data is available on the web. It is difficult to retrieve precise information from the current web. Semantic web is a solution for semantic interoperability. There are different existing ontology in the education domain. Nowadays, the Industry-Institute gap is increasing due to various reasons. These gaps can be bridged after finding gaps by planning and conducting different activities for students. It will automatically reduce the unemployment rate. Industry and Institute both are equally responsible to develop quality students. In this paper, cross domain (Industry domain and Institute domain) ontology based semantic models are developed to bridge the institute-industry gap using Protégé 5.5.0 editor. The classes and sub classes of Industry-Institute ontology are designed with the help of domain experts. Then, object property and data property are defined to enhance ontology. Next, the result of ontology is validated using HermiT reasoner, DL query and SPARQL query. The graphical representation of ontology is shown by using OntoGraf, OWLViz and VOWL plugin.

Keywords: *Ontology, Semantic Web, protégé 5.5.0, Cross Domain, Data Annotation, Data Filtering*

1. INTRODUCTION

Institute can provide everything to students as per industry requirement. The existing education domain ontology is mostly implemented for universities, courses, and libraries. In this paper, we have constructed cross domain ontology to bridge the gap between industry and institute. Ontology development takes a lot of effort and time. We surveyed a number of papers on the education domain. There is no ontology which bridges the industry-institute gap.

Why is the Industry-Institute gap increasing day by day?

The main reasons are as follows.

1. Students' interest is changing.
2. Students are not enjoying old teaching-learning methodologies.
3. Less interaction with industry.

There is a necessity to fill these increasing gaps, else it will increase the unemployment rate in the near future. Unemployment Rate is increasing day by day [1] [2]. It does not mean the jobs are not available. A number of jobs are available but students do not have deep knowledge to work on real time projects.

Actions taken to bridge gap:

1. Need to design more practical based curriculum.
2. Conduct more technical and non-technical activity.
3. Teaching and learning must be enjoyable and joyful.
4. Faculty skills development program - by organizing workshops and industry training.

The major contribution of paper are as follows:

1. Develop Institute Ontology
2. Develop Industry Ontology
3. Merged Institute and Industry ontology using Protégé 5.5.0
4. Execute DL queries and SPARQL Queries
5. Validate Ontology (OntoGraph, VOWL, OntoViz)

Progress In Electromagnetics Research M, Vol. 114, 233–243, 2022

Compact Planar Electromagnetic Bandgap Structure for Signal and Power Integrity Improvement in High-Speed Circuits

Manisha R. Bansode[1, *] and Surendra S. Rathod[2]

Abstract—This paper introduces and validates a compact two-dimensional Electromagnetic Bandgap (EBG) structure for the improvement of signal integrity (SI) and power integrity (PI) by suppressing Simultaneous Switching Noise (SSN). SSN bandwidth can be increased by using the proposed T bridge compact planar structure. The proposed structure is simulated using Ansys HFSS Software. Simulated and measured results by Vector Network Analyzer provide 3.13 GHz to 11.40 GHz frequency bandgap with good mitigation of SSN at −30 dB noise suppression reference. It will almost cover S, C, and X bands from electromagnetic frequency spectrum. This will be useful for satellite and terrestrial communication and radar communication applications. The proposed structure analyzes signal integrity issues using eye diagram in MATLAB and power integrity in HFSS with input impedance respectively. The main purpose of this work is to provide a compact structure to improve signal and power integrity by the suppression of power/ground noise. Comparative study is also performed with the proposed structure and reference board with similar dimensions.

1. INTRODUCTION

Nowadays, the high-speed design in GHz band has a main focus on signal integrity and power integrity of a system. Work gets more difficult for electronics business as a result of decreased voltage levels, cost efficiency, miniature size, and higher data rate of a system. System operates as per decided design guidelines for lower frequency, but system issues including reflection, ringing and crosstalk, and other electromagnetic interferences will become worse as frequency rises. Due to clock frequency and data rate rising continuously, simultaneous switching noise (SSN), a major problem, is growing for higher frequency applications. Due to cavity mode resonance on the parallel plate waveguide power planes of the high-speed printed circuit boards and the Power Distribution Network (PDN), simultaneous switching noise (SSN) is an undesired voltage fluctuation emerging [1–3]. There are signal integrity and power integrity problems that arise due to resonance modes in power and ground planes. Other significant problems in high-speed circuits and PCBs include power integrity, electromagnetic interference, and signal integrity. Researchers found effects of these issues in both time and voltage domain. Therefore, It is necessary to maintain signal and power integrity in electronics systems. By changing the timing of the system, we can reduce the impact of noise [4–6].

For the suppression of electromagnetic interference, many solutions are found like the use of decoupling capacitors, splitting planes, differential interconnection, and shorting vias. Typically, decoupling capacitors are positioned on top of the PCB. To separate high frequency oscillations from the power plane, capacitors are positioned close to the noise source. At specified frequencies, it establishes a low impedance path between the planes.

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Dual Degree of Freedom Solar Tracking System

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ABSTRACT—Photovoltaic cells are designed to receive incident sun rays and convert it into equivalent electrical energy. But these photovoltaic panels are not that efficient as they are freezed only at a particular angle. This inefficiency can be reduced by designing a solar tracking system which automates its position as per the sun's movement. The goal of this proposed design is to get maximum power output, a dual-degree of freedom solar tracker is designed. The tracker actively traces the sun motion and automatically changes its position accordingly to increase and maximize the power output.

Index Terms—Solar Cell, Panel, Dual Axis, Two DoF, LDR, Maximum Power, Servo Motor.

I. INTRODUCTION

The incident sun rays can be used to overcome the electrical energy crisis generated by the scarcity of Fossil fuel resources. Solar energy is free and everywhere. Due to the decreasing of solar photovoltaic energy cost, it's superior in the renewable energy sources and widely utilized in many countries. One of the most widely used alternative route in the frame of renewable energy domains or sources is Solar Energy. Solar power is a very exorbitant, unlimited source of energy. The energy from the sun received by the earth is estimated to (1.9×1011) Mega Watts, which corresponds to many thousands of times more than the current usage rate on the earth from all of all energy sources available commercially. Issues concerned with the utilization of solar power is that its accessibility varies broadly with space and time. To solve these issues, the solar panel should be installed such that it always intercepts maximal intensity of light it can. It has been observed since many years of experimentation that the efficiency of the solar panel is around 15-20% which is not meeting the desired load requirement. This problem can be rectified by means of efficient solar trackers. Solar tracking approaches can be enforced by using single degree of freedom or dual degree of freedom for better accuracy and high efficiency. In general, the Single-Degree of Freedom tracker with one degree of freedom follows the Sun's motion from the geographical east to geographical west during daytime. If Dual-Degree of Freedom tracker is considered, then it will follow the elevation angle of the Sun [1].

II. METHODOLOGY

As mentioned before, this project is to visualize and analyse the benchmark and performance of dual- Degree of Freedom solar tracking system. There are three prime components: inputs from the LDRs, the controller to control the entire assembly and the output to carry out actuation. The inputs are taken from LDRs to detect the intensity of sun-rays, the Arduino is acting as the controller to take necessary action in setting angles and, the servo motor as the actuator to set the desired angles fed by controller output. The overall system is presented in Figure 2. In this project, the prime controller, Arduino accepts analogue input from LDRs and digitized the input signal data using analogue-to-digital (A-D) converter. Then the controller feeds the desired angles to servo motor in order to carry out the actual actuation determining the motion of the solar panel [3].

ADVANCES IN MATERIALS AND PROCESSING TECHNOLOGIES
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Analytical modeling and numerical investigation of a variable width piezoresistive multilayer polymer micro-cantilever air flow sensor

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ABSTRACT

Measurement of air flow is a vital concern in diverse domains. Piezoresistive micro-cantilever-based air flow sensing is a promising research domain within MEMS technology due to its several advantages. Displacement, bending stress, sensitivity, resolution and stability are significant challenges in the design of this sensor. These parameters are also affected by the cantilever profile and its materials. Hence, the purpose of this study is to investigate the performance of a piezoresistive multilayer polymer variable width profile micro-cantilever structure and propose its materials, fabrication process flow, optimisation of geometric dimensions, piezoresistors placement and piezoresistors circuit connections to address challenges in the design. Methodical steps towards design optimisation of the proposed sensor under the applied pressure due to its interaction with air flow comprise preliminary approximation of geometrical dimensions through analytical models developed using Conjugate Beam and Rayleigh methods, subsequently, 3D modelling numerical simulation using the COMSOL Multiphysics FEM tool. It is observed that numerical simulation results agree with the results of analytical modelling. It is concluded that the proposed study contributes to the development of accurate analytical models and design optimisation of a low-cost and stable micro-cantilever-based air flow sensor that performs yet equally well compared to previously published designs in terms of sensitivity.

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Accepted 9 May 2022

KEYWORDS
Bending stress; cantilever; deflection; optimisation; variable width

1. Introduction

A sensor translates any physical phenomenon to an electrical quantity [1]. Air flow measurement is a vital concern in diverse domains such as weather condition monitoring in the meteorological department and estimating power generation using wind, aeronautical, construction, transportation, medical instrumentation and process control [2]. Mechanical and acoustic anemometers use mechanical apparatus and Doppler shift of frequency phenomenon, respectively, for wind speed sensing. The drawback of these traditional sensors is the large response time due to mechanical components.

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RESEARCH ARTICLE

Privacy-Preserving Mechanism to Secure IoT-Enabled Smart Healthcare System in the Wireless Body Area Network

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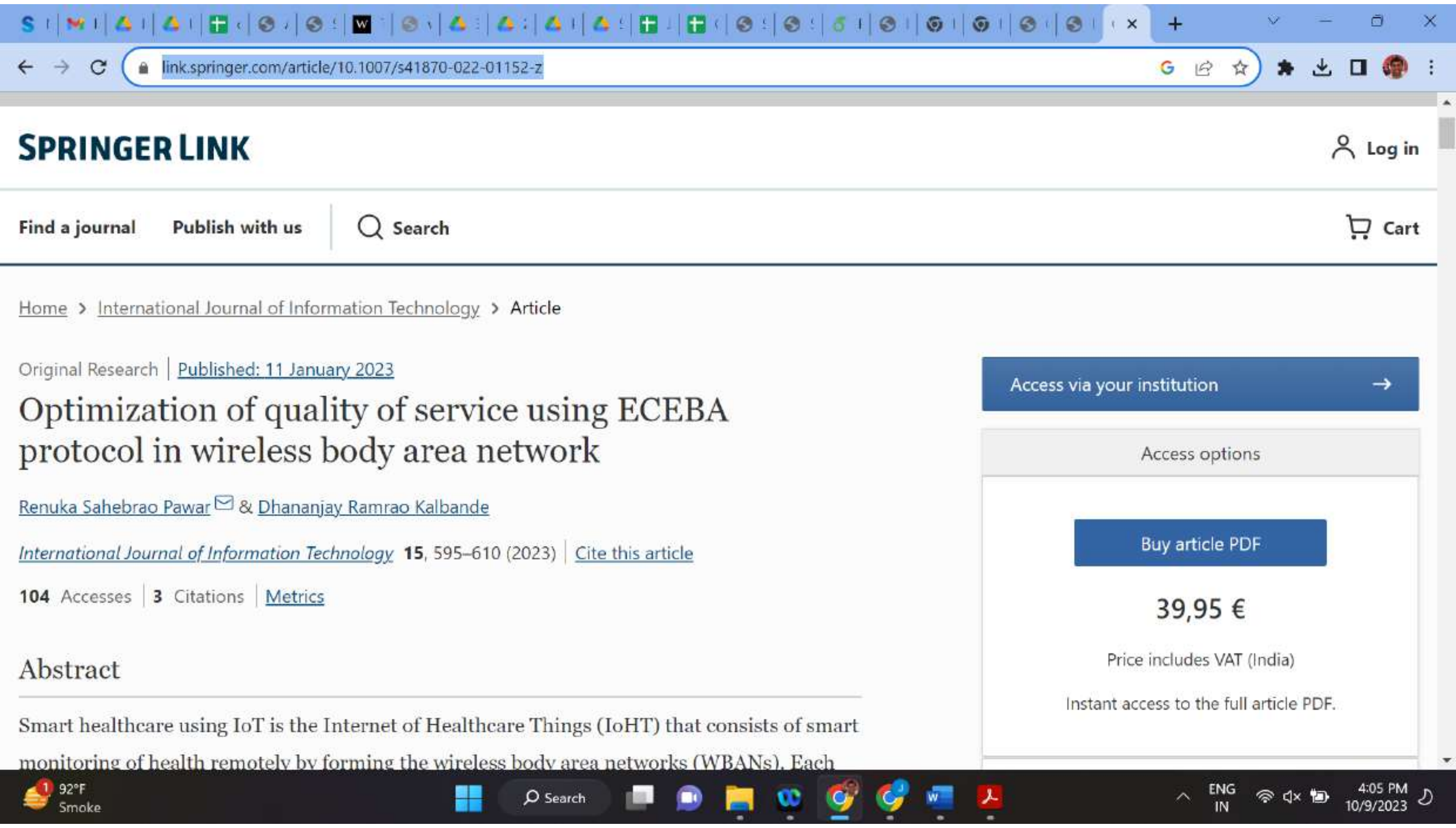
Abstract – The IoT has been a subclass of Industry 4.0 standards that is under research from the perspective of quality of service (QoS) & security. Due to the pandemic situations like novel coronavirus smart healthcare monitoring gained growing interest in detection. In IoT data is communicated from Intra WBAN (Wireless Body Area Network) to inter-WBAN and then beyond WBAN. While transferring data from one layer to the other end-to-end data privacy is the challenge to focus on. The privacy-preserving of patients' sensitive data is difficult due to their open nature and resource-constrained sensor nodes. The proposed research design based on routing protocols achieves the patient's sensitive data privacy preservation along with minimum computation efforts and energy consumption. The proposed model is Secure Communication-Elliptic Curve Cryptography (SCECC) WBAN-assisted networks in presence of attackers is evaluated using NS2. The proposed privacy preservation algorithm uses efficient cryptographic solutions using hash, digital signature, and the optimization of the network.

Index Terms – Authentication, Cryptography, ECC, Encryption, Wireless Body Area Network, Routing Protocol.

1. INTRODUCTION

The smart healthcare system using the IoT enables hospitals to monitor their patient's health from any place in the world periodically [1]. Smart healthcare is mainly based upon Wireless Sensor Networks (WSNs) under which the patients have been equipped with body sensors & hence acts as a node in the network. Such networks are also defined as Wireless Body Area Network (WBAN). Because of the progressions in Micro-electromechanical systems (MEMS) and wireless correspondence advances, WBAN has experienced a specialized impact in the most recent decade. There are wide assortments of security assaults that are conceivable because of the wireless idea of the correspondence medium. Consequently, security must be given in the application layer just as in the network layer [2-4]. Through information security, the assurance of information from unapproved clients while information is being put away and moved benefits the people to control the assortment and utilization of individual information about them [5-7]. The security instruments for shrewd medicinal services or WBANs incorporate (1) Data stockpiling: Confidentiality, Integrity confirmation, Dependability, (2) Data get to Access control, Accountability, Revocability, Non-denial, and (3) Other: Authentication and Availability. Furthermore, the WBANs data will be available almost everywhere, anytime. Inescapably, & such data has been beggarly sensitive & problematic hence an attacker may insert malicious nodes under the network for leaking the body sensor's medical data (&/or involves false reports) except permission regards the network owner [8-11]. There are several access control policies introduced, however, selecting the appropriate approach to achieve privacy preservation is a challenging task considering the limited capabilities of sensor devices (constrained battery, low memory, less bandwidth, & lower bandwidth). How to address the privacy-preserving requirement of smart healthcare systems using WBAN-assisted IoT is a big forthcoming issue for many researchers. Designing the access control policies to address these issues needs to satisfy the requirements of energy efficiency along with strong security without disclosing the sensor's identity. The reason regards this research has been for describing designing & evaluation of access control policies-based routing schemes to achieve privacy preservation in the smart healthcare system. Along this connection, we first design the system model of a smart healthcare system using WBAN-assisted IoT in the presence of malicious nodes in the network. Secondly, the two existing access control protocols such as ACP and CDHSC] are described along the



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Original Research | Published: 11 January 2023

Optimization of quality of service using ECEBA protocol in wireless body area network

Renuka Sahebrao Pawar & Dhananjay Ramrao Kalbande

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Abstract

Smart healthcare using IoT is the Internet of Healthcare Things (IoHT) that consists of smart monitoring of health remotely by forming the wireless body area networks (WBANs). Each

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QOE Improvement for Dynamic Adaptive Streaming of Multimedia in LTE Cellular Network Using Cross-layer Communication

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Abstract: This Research focuses on cross layer approach to enhance the User Experience and Quality of Service of video streaming in Long Term Evolution mobile network. During run time channel quality index is observed. Application layer requirement is fulfilled by changing modulation techniques as well as dynamic allocation of resources. This paper proposes a algorithm which improves the End-to-End Delay, Peak Signal to Noise Ratio of transmitted video over Mobile Network. Experimental results show the improvement in end to end delay by 89% and Peak Signal to Noise Ratio by 8%. Simulation in Network Simulator-3 provide credible evidence that this proposed cross-layer algorithm outperforms between earlier algorithm in providing better Quality Of Experience for real time, adaptive video streaming over Long Term Evolution.

Index Terms: Cross Layer, Long Term Evolution, Mobile Networks, Mean Opinion Score, Peak Signal to Noise Ratio, Quality Of Experience, Quality Of Service, Reference Signal Received Power, Structure Similarity Index Method.

1. Introduction

The Video streaming in the cellular Network is growing and expanding services due to the emergence of multimedia based applications. The video industry has adopted dynamic streaming over Hyper Text Transfer Protocol(HTTP), to provide smooth viewing experience in varying wireless environment. However there are several challenges to develop robust streaming solutions in mobile network. In streaming application the video content is sent in compressed form over the internet and at the receiver it is decoded and played in real time. The number of parameters need to be considered during live video streaming in wireless environment. Some of the factors that influence the video streaming are delay, packet loss, jitter and compression techniques. Internet traffic is time dependent, show the quality of service (QOS) in heterogeneous networks becomes challenge. However QOS is an end-to-end key issue for communication link quality measure. In wireless channel link conditions are fluctuating, which demand for changing bandwidth that requires adaptive video streaming for the effective and higher quality of experience (QOE). Further, it may create video freeze at display in the user equipment due to more weight for the video frame to buffer. [1]

The LTE cellular services have been deployed to match the growing traffic demand for video streaming, with a peak downloading bit rate of 300 mbps, almost 10 times more than 3G. However, user dependent quality of experience (QOE) remains unsatisfactory. A recent world-wide measurement survey [2] reveals that, even wide LTE coverage in the regions, will boost the video quality by 20% over 3G network on the other hand, the average stalling time remains 7.5 to 11.3 seconds for each minutes mobile video playback. [3]

These two effects are seemingly antithetical: the video streaming applications are sorely under utilize the LTE bandwidth, but stalling happens due to bandwidth overestimation. However, based on Reference Signal Received Power (RSRP) measurements study, we found a single root cause behind is the inability of the streaming application to track the network bandwidth which is affected by down link traffic dynamics at the E-nodes (Base Stations) of LTE.

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Ultrasound Image Enhancement using Super Resolution

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ARTICLE INFO

Keywords:
Medical image denoising
Hybrid filters
Early fibroid detection
Super-resolution convolutional neural network

ABSTRACT

Within today's day and age, amongst women who fall under the bracket of being of reproductive age, when discussion about Gynecologic tumors comes forward, Uterine Fibroids are found to be abundant. Since early detection for tumors is statutory for ensuring recovery, ultrasound detection remains prominent for being one of the most convenient Image-capturing methodologies. Although given the convenience of the method, the quality of the images could be deemed sub-optimal, a need to enhance the image arises for successful detection by appropriate algorithms. This all facilitates the development of a technique with a harmonious combination of filters and Super-Resolution Convolutional neural network (SRCNN).

Super Resolution technique in itself is altered to make it feasible for monochromatic medical images, which essentially highlights the contagious aspect, i.e., the tumor region of the image. Thus, firstly the captured noisy image is denoised by making use of Spatial filters, which ensures denoising while preserving the essential edges. Then, this output of the filtered image is treated as an input for the Super-resolution technique, as described above. Within the said technique, multiple low-resolution images are combined in Super-Resolution to regenerate a target input image with higher resolution and quality. Furthermore, Subjective, quantitative, and visual execution evaluations were conducted using Peak Signal-to-Noise-Ratio (PSNR) for quantitative and Universal Quality Index (UQI) and structural similarity index metric (SSIM). Upon completion, these results of comparing the quality of super-resolved images on various data sets are provided in numerical form.

These comparisons state that the experimental results have shown a substantial improvement in the results by the fusion technique of hybrid filters with Super Resolution convolutional neural network, for instance, the PSNR value achieved is in the range of 40% to 50%. As a result, the proposed technique has demonstrated outstanding performance in all cases, as evidenced by the findings that the proposed algorithms are robust and ultrasound images taken under various lighting conditions and parameters work efficiently and without egregious errors.

The paper is organized as follows: Section 1 gives the introduction for requirement of ultrasound medical image enhancement. Area 2 depicts the proposed strategy for a sorting and Super Resolution fusion technique. Area 3 describes the assessment databases as well as the experimental setting. We look at the trial results and have a discussion in region 4. The paper ends in zone 5 by showing that the new system outperforms the existing one.

1. Introduction

After considering everything in hand, the prospect of extricating the fibroid from a low-resolution ultrasound scanned uterus image, complemented by a high degree of commotion, presence of imaging artifacts, scale, position, and low contrast boundaries, comes across as tedious and cumbersome of an affair. Given the above fact, X-ray, Ultrasonography, Magnetic Resonance Imaging, and Computer Tomography (CT scan) are some of the diagnostic procedures utilized for uterine fibroid, where MRI produces an appropriately high-quality image but at a significantly high cost. Therefore, one might look towards other techniques, such as X-ray and CT scans, which are comparatively more affordable, but then they create certain health hazards at soft sensitive tissues, eliminating even these methods as a feasible way of detection. As a result, improving ultrasound images with the combination strategy of filters and super-resolution methods would not only offer a cost-effective approach, but it would also be deemed a reliable, safe, and healthy approach in clinical imaging for the detection of the fibroid.

Although current data from imaging sensors on different frameworks consolidates defects, diagnostic imaging strategies facilitate the interpretation of collected medical images. Such images must be subjected to long-term processing in order to resolve inconsistencies and thereby

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Helping Hand: Metro Travel Assistance App for Elderly and Differently Abled People

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Abstract

Abstract— As per Census 2011 there are 59,392 persons with disabilities in Maharashtra and it is 2.63 % of total population of Maharashtra State. An aging population increases the demand for health services and assistance services. The problem of assisting and helping in day to day needs for the elderly has increasingly become one of the focuses of social concerns. Because of the popularization of mobile devices and wearable devices like smartwatches, using these devices to assist the care of the elderly has become a feasible solution. Differently abled or elderly people can navigate through the places that they have already been in with rather ease. But if they have to go to a new, unfamiliar place, that is where they face some issues. In this paper, we developed an elderly assistant system based on mobile and wearable devices, which includes two apps i.e., the Guardian side and the Elderly side. The guardian side remotely helps the elderly in case of emergency SOS request; the Elderly side assists the traveling needs and emergency needs of the elderly/differently abled people. The Developed assistance app provides real time travel assistance in metro stations, and SOS support in emergency situations. The aim of this assistance app is to provide a trustworthy system where the needy can travel safely and get the required assistance.

Keywords: mobile, elderly, public transport, helping, travel

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Mood Detection using Sentiment Analysis

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ABSTRACT
Since this pandemic has taken place the world is not in a great condition physically and mentally. People are witnessing all kinds of unexpected circumstances. Under such circumstances, people tend to break down. Mental health is like a catalyst to physical health; it can either boost up or tank the entire recovery mechanism of a person. Everyone should have a healthy mindset toward life. Mood Trend is a system which aids users' mindset or mood with small amounts of positive reinforcements. In this way, we make a significant impact on users' lives in the long run. The website will detect the sentiment behind the user's words and provide content to enhance their mood. The user can view this content material to lighten his/her temper or to control stress. Mood Trend also has an option where the users can upload a picture of themselves and the system will detect sentiment.

Keywords
Sentiment Analysis

1. INTRODUCTION
The way you feel at a specific time is how mood is defined. Mood is a mixture of feelings and emotions as we go through our days. A mood is a semi-persistent mental + physical + emotional state. Sentiment is a thought, opinion, or idea based on a feeling about a situation, or a way of thinking about something. Analysis is a thorough investigation of a complex object in order to comprehend its nature or establish its key characteristics. To analyze a topic or concept, you must disassemble it into its constituent elements in order to inspect and comprehend it, and then reassemble those parts in a form that makes sense to you

Sentiment analysis is a natural language processing technique used to determine whether data is positive, negative or neutral.Sentiment analysis is commonly used on textual data to help businesses analyze brand and product sentiment in customer feedback and better understand customer needs. Businesses frequently use it to detect sentiment in social data, assess brand reputation, and comprehend customers. Sentiment analysis is critical since it allows firms to quickly assess their customers' overall opinions. By automatically sorting the sentiment behind reviews, social media conversations, and more, you'll make faster and more accurate decisions.90 percent of the world's data is unstructured, or unorganized, according to estimates.

Mood Trend will basically detect the users' mood over a period of time. A person shares his/her thoughts via a medium, it can be a closed one or a diary. Similarly, the website provides an option for users to submit their ideas on a daily basis. The thoughts are stored inside the database and could be used for sentiment analysis. Basic sentiment analysis of text follows a straightforward process: 1. Break each text document down into its component parts like sentences, phrases, tokens and parts of speech. This is known as tokenization. 2. Then we Identify each sentiment-bearing phrase and component. 3. And finally we Assign a sentiment score to each phrase and component.

For identifying sentiments, there is something called a sentiment library. An efficient sentiment analysis system depends upon the right sentiment library to efficiently detect sentiments and scores in phrases or terms. Sentiment libraries are composed through a group of dictionaries along with adjectives and terms that have been manually scored beforehand. This scoring technique ought to be accomplished cautiously in order that the sentiment analysis system can differentiate among phrases like 'bad' and 'horrible', holding 'horrible' a more negative meaning. Aside from that, a multilingual sentiment evaluation engine will come with libraries for each language supported. Once those dictionaries are prepared, a sequence of policies must be written inside the software program so the system is capable of detecting the sentiment expressed closer to a selected topic, primarily based totally on its proximity to different tremendous and bad words.

2. LITERATURE REVIEW
Many research papers are published on sentiment analysis. There are many domains and fields where sentiment analysis is applied and implemented. A few of them are mentioned below:

1. MELD: A Multimodal Multi-Party Dataset for Emotion Recognition [1]

The authors of this study address the growing popularity of research on conversational emotion recognition. As well, they state there is a lack of emotional conversational databases in the field.

2. Large-Scale Sentiment Analysis for News and Blogs [2]

Newspapers and blogs express the opinion of news entities like people, places, things while reporting on recent events. This paper demonstrates a system that assigns positive or negative scores to each different entity in a text corpus.

The Need of Explainability to Enhance the Trustworthiness of Decision-Making System

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Abstract— **The development of effective and highly accurate models using Machine learning and deep learning in the past decade has boosted interest in the field of artificial intelligence. But the escalating acceptance of AI has also emphasized some of the critical problems of the field, including the "black box problem," the challenge of getting sense of the way intricate machine learning algorithms which helps in making the decisions. To enhanced openness and belief in AI, now a day research in Explainable Artificial Intelligence (XAI) must need. This is essential as AI is used in many sensitive domains. This paper shows the case study focusing on the False-Positive Credit Card evaluation. Fraud costs 15 times more in lost income than true fraud. This model determines a fraud score, and bank can easily work on the decisions.**

Keywords— **XAI, black-AI, ML**

I. Introduction

AI is gradually playing an essential role in enhancing our day-to-day experiences. Moreover, with Areas like hiring, lending, criminal justice, healthcare, and education there is a huge rise of solutions which are based on AI. The leading role of AI based models in many fields has led to a rising concern about probable bias in these models, so there is a demand for openness and explainability of the model. Eexplainability is an essential for enhancing the trust and acceptance of AI systems in high risks domains where the trustworthiness and security are the must needed factors. Many research scholars have focused on XAI term which help them to create a better trust and explainable models. The challenges which is faced by the research scholars include defining explainability of the model, Reframing the tasks for better understanding of the model, creating new outcomes for tasks and discovering the parameters for assessing the model's performance.

1. Literature Analysis:

AI system can be further extended giving more support for intelligent support. Explainable machine learning algorithms for credit scoring tackled the problem of black box in ML. They also compare credit risk modelling with new advanced modern statistical tools by developments in interpretability and explainability of artificial intelligence algorithms [1]. AI generates new prospects that create a remarkable revolution in the overall financial systems. It causes the fast growth of big data opportunities. XAI upgraded customers' satisfaction level and acceptance [2]. Electronic-commerce improves their services' proficiency and superiority. Sometime, AI suggest the preference which doesn't meet the customers' requirements. Bi-step processes which are explanandum unfolding the root for an event. Interactive manner is used for conveying knowledge between explainer and explainee in social process. [3] XAI means explanation for mapping with demands along with the developers and researchers. For the better accepting models and increased use of AI, the performance betterment in ML model created[5] [6]. Two opposite methodologies will generate more clear reasonable systems to develop acceptance level and belief the models [7]: It handles interpretability and explainability. XAI promotes building the trust and manages the new generation of AI [8]. The decision tree algorithm for the malevolent node detection by using a publicly accessible KDD dataset. They presented three crucial operations in this dataset for features ranking, rule extraction of decision tree, and comparison with the state-of-the-art algorithms.

2. Motivation of XAI from Research Perspective

Machine learning work to acquire expertise to generate the prediction. But common techniques produce opaque content where it is difficult to predict the behavior. Model need to specify the trust for the decision made by the system.

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PHYSICAL REVIEW A (/PRA/)
covering atomic, molecular, and optical physics and quantum information

Electron emission from water vapor under the impact of 250-keV protons

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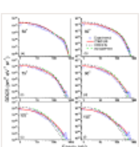
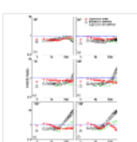
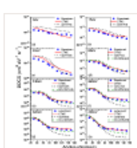
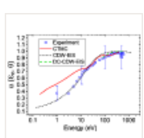
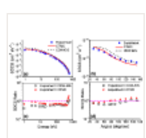
ABSTRACT

Absolute double differential cross sections (DDCS) of low-energy electron emission from water molecule were measured upon collisions with 250-keV protons for emission angles between 30–150 degrees. The electrons having energies between 1 and 600 eV were detected by using hemispherical electrostatic analyzer. The single differential (SDCS) and total cross section (TCS) were calculated by integrating the DDCS and SDCS, respectively. The measured DDCS, SDCS, and TCS were compared with the classical trajectory Monte Carlo (CTMC) model, using a dynamical approach in which a time-dependent screening is considered. The continuum-distorted-wave eikonal-initial state (CDW-EIS) theoretical model has also been used to explain the energy and angular distribution data. The angular distribution of the DDCS are very well reproduced by the CTMC approach. The TCS calculated by the CTMC model matches better with the measured values as compared to the CDW-EIS estimation. The forward-backward angular asymmetry parameter was also estimated to test the validity of the state-of-the-art theoretical models used. Finally the recently developed CDW-EIS calculations considering a residual target dynamic charge (DC-CDW-EIS) is shown to provide an improved agreement with the experiments compared to the CDW-EIS. The present data and interpretation should provide inputs towards energy loss calculations required for the radiobiology involved in the hadron therapy of cancer.

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